

PROJECT DESIGN PHASE-II

Solution Requirements Document (Functional & Non-Functional)

Date	4 August 2026	Project Name	Online Complaint Registration System (MERN Stack)
Maximum Marks	4 Marks		

Overview: This document specifies the comprehensive functional and non-functional requirements for the Online Complaint Registration platform. The system aims to bridge the gap between citizens and public/organizational service departments by offering a seamless online complaint filing, tracking, and resolution management platform.

1. FUNCTIONAL REQUIREMENTS (FR)

The functional requirements define the primary operational capabilities, core system features, and user interaction workflows.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	- Registration through Custom Form (Name, Email, Phone, Password) - Registration through Google OAuth Single Sign-On (SSO) - Mobile OTP-based Registration for citizens without an email address
FR-2	User Authentication & Confirmation	- Account Confirmation via Secure Activation Link sent to Email - Identity Verification via 6-Digit Time-based Mobile OTP - Two-Factor Authentication (2FA) for Officers and Admins
FR-3	Complaint Filing & Categorization	- File Complaint by Department/Category, Description, and Geo-tagged Location - Multi-file Upload for Photo/Video Evidence Attachment - Auto-suggestion of Category based on Complaint Description Keywords
FR-4	Complaint Tracking & Management	- Real-time Complaint Status Dashboard for Citizens and Officers - Officer Assignment & Reassignment Workflow by Department - Automated SLA-based Escalation for Overdue Complaints
FR-5	Notifications & Communication	- Automated SMS & Email Status Update Notifications - In-app Notification Center for Status Changes - Citizen Feedback & Satisfaction Rating Submission after Resolution
FR-6	Admin Reporting & Analytics	- Department-wise Complaint Volume & Resolution Analytics Dashboard - Officer Performance Tracking (Resolution Time, Backlog) - Exportable Reports (PDF/Excel) for Audits and Reviews

2. NON-FUNCTIONAL REQUIREMENTS (NFR)

The non-functional requirements specify quality criteria, performance standards, regulatory compliance, and system constraints.

NFR No.	Non-Functional Requirement	Description & Acceptance Criteria
NFR-1	Usability	Intuitive, mobile-responsive user interface ensuring citizens can complete a complaint submission within 3 clicks. Meets WCAG 2.1 Level AA accessibility standards.
NFR-2	Security	End-to-end AES-256 encryption for sensitive complaint/citizen data at rest and TLS 1.3 in transit. Strict compliance with local Data Protection and RTI-adjacent privacy regulations.

NFR-3	Reliability	High-reliability complaint engine achieving 99.9% successful submission rate with zero data-loss occurrences through transactional database writes.
NFR-4	Performance	System response time under 1.5 seconds for complaint status queries. Supports 500 requests per second (RPS) under peak operational load without performance degradation.
NFR-5	Availability	99.95% cloud service availability with automated multi-region failover. Maintenance windows scheduled strictly during off-peak hours (02:00 AM-04:00 AM IST).
NFR-6	Scalability	Horizontal autoscaling architecture capable of handling concurrent traffic bursts up to 25,000 active users and 100,000 daily complaint status queries.
NFR-7	Maintainability	Modular microservices architecture structured with containerized services (Docker/Kubernetes) allowing independent deployments and continuous updates.
NFR-8	Data Integrity	Automated daily database backups with 30-day retention policies and automated point-in-time recovery (PITR) mechanisms with an RPO < 5 minutes.

System Compliance & Compliance Note: All functional and non-functional requirements detailed above adhere to the Phase-II project evaluation criteria. Implementation phase verification will be performed via automated end-to-end integration tests and third-party security audits.

Prepared By: Project Development Team **Approved By:** Project Evaluator / Guide